

Sasikanth Vadde

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PROFESSIONAL SUMMARY

Chemical & Energy Engineer (M.Sc.) specializing in electrochemical and multiphysics modelling of alkaline water electrolysis (AWE). Two years at DLR (German Aerospace Center), progressing from intern to Research Associate and taking a single-cell **Dymola/Modelica** model to a validated high-pressure stack digital twin. CFD in **OpenFOAM**; Python/PyTorch for physics-informed ML.

KEY SKILLS

Alkaline Water Electrolysis (AWE)

Dymola/Modelica Digital Twins

Electrochemical & Two-Phase Modelling

Model Validation with Test-Rig Data

Parameter Identification

CFD & Conjugate Heat Transfer (OpenFOAM)

Multi-Loop PID Process Control

Python, PyTorch & Physics-Informed ML

WORK EXPERIENCE

DLR (German Aerospace Center), Institute of Engineering Thermodynamics 09/2023-12/2025

Intern, Working Student, Master's Thesis Student, Research Associate

Stuttgart

- Value added: Progressed from intern to Research Associate, taking a single-cell AWE model to a validated high-pressure stack **digital twin** used for system-level KPI assessment.
- Modelling: To simulate transient behaviour, built a 1D single-cell AWE model in **Dymola/Modelica** from published electrochemical, two-phase and thermal correlations, then extended it to a **high-pressure stack model**.
- Modelling: To improve Henry's-law solubility predictions under pressure, added a **fugacity correction** for non-ideal gas behaviour and validated the de Groot supersaturation correlation across **1-20 bar and 60-80°C**.
- Validation: To assess low-load safety risk from reduced gas dilution, validated the H2-in-O2 gas-purity model against test-rig data at three electrolyte flow rates in mixed and separated circulation, reaching **under 5% deviation**.
- Calibration: To match measured behaviour over the full operating range, fitted kinetic and gas-crossover parameters to DLR test-rig data and derived an **empirical current-density-based** diffusion correlation.
- Thermal: To calibrate the Nusselt-number correlations of the thermal stack model, ran conjugate-heat-transfer CFD in **OpenFOAM** (chtMultiRegionFoam) of endplate cooling-channel flow.
- Control: To keep gas purity within safety limits under fluctuating renewable load, developed **multi-loop PID control strategies**; generated ARMA-based wind and PV load profiles in Python to drive the transient simulations.
- Tools: Dymola/Modelica, OpenFOAM, Python

Otto von Guericke University Magdeburg 06/2023-08/2023

Working Student (HiWi), Chair of Mechanical Process Engineering (Prof. Berend van Wachem)

Magdeburg

- Image analysis: To calibrate a drying-process model, built a **Python/OpenCV** bubble-tracking pipeline for fluidized-bed images.
- Experiments: Ran fluidized-bed experiments with varying inlet pressure and sinusoidal airflow on the test rig.
- Tools: Python, OpenCV

GMR Group (Hyderabad and Delhi airports, Bajoli Holi hydro power plant)

2019-03/2021

Graduate Engineer Trainee, then Junior Engineer, Airside Operations

India

- Commissioning: Contributed to **HVAC, chiller and fire-system commissioning** during a rotation across airport and hydro power plant sites.

Hindustan Aeronautics Limited (HAL)

11/2018-04/2019

Full-semester Internship

Koraput, India

- Quality: To assess process stability, analysed manufacturing, inspection and fatigue-testing sequences for AL-31FP turbine blades using **statistical process control** (X-bar and S charts).

PROJECTS

Physics-Informed Neural Network for Battery State-of-Health since 02/2026

Combined a data-driven model with an electrochemical Single-Particle-Model ageing model for large-format LFP cells; shared MLP with learnable cell embeddings and a differentiable physics pipeline; **zero-shot capacity prediction on held-out cells** and forecasting from 70% of a cell's history, evaluated with RMSE/R²/MAPE.

PyTorch

Reversible Solid Oxide Cell (R-SOC/SOEC) Process Simulation

Process simulation of a reversible solid oxide cell with custom Python thermodynamic and control scripting.

DWSIM, Python

Exergoeconomic Optimization of a 210 MW ISCC Power Plant

Iterative optimization over 5 decision variables: **-6.2% natural gas use**, LCOE_{ex} reduced from 93.41 to **90.07 EUR/MWh**.

Total Revenue Requirement method

LCA of Lithium Battery Recycling

Aspen Plus recycling process model (dismantling, leaching, Li recovery) with mass and energy balances, coupled with a life-cycle assessment in openLCA (ReCiPe midpoint).

Aspen Plus, openLCA

EDUCATION

M.Sc. Chemical and Energy Engineering Otto von Guericke University Magdeburg

04/2021-10/2025

Magdeburg

Grade 1.9. Thesis: Transient Modelling of an Alkaline Water Electrolyzer for Validating a Two-Phase Electrolyte Model (MPI Magdeburg + DLR Stuttgart). Focus: CFD, Process Control & Systems Engineering, Multiphase Flows.

B.Tech Mechanical Engineering GMR Institute of Technology

07/2015-04/2019

Rajam, India

Focus: Thermodynamics & Heat Transfer, Fluid Mechanics, FEM, Power Plant Engineering & Energy Systems.

TECHNICAL SKILLS & LANGUAGES

Simulation: Dymola/Modelica, MATLAB/Simulink, OpenFOAM, STAR-CCM+, ANSYS, COMSOL, Aspen Plus, DWSIM

Modelling: electrochemical kinetics, two-phase/multiphase flow, heat & mass transfer, coupled multiphysics

Process & control: PID & multi-loop control, P&ID/PFD, PLC (IEC 61131-3), measurement-data & root-cause analysis

Programming & data: Python (PyTorch, scikit-learn, NumPy, Pandas), C++, MATLAB, Git, openLCA, AutoCAD

Languages: German (B1, working toward B2), English (C1/C2), Telugu (native), Hindi (native-level)